

Figure Reproduction Instructions for the Manuscript titled "Data-driven model order reduction for wave propagation in materials with damage and nonlinearities"

This document outlines the step-by-step procedure to reproduce all numerical figures in our manuscript.

Software Environment & Versions

The computations were performed using the following exact software versions:

Software / Package	Version
MATLAB	R2023b
Python	3.10.12
TensorFlow	2.18.0
NumPy	1.26.4
SciPy	1.12.0

Prerequisite Note

The folder `Matlab_Routines` must be included in the MATLAB search path so that custom formatting settings (fonts, legends, etc.) are applied correctly and core dependencies for the reduction methods are loaded. You can place the folder directly inside your working directory or execute the following command at the start of your MATLAB session:

```
addpath(genpath('Matlab_Routines'));
```

In some MATLAB versions you might need to perform the following command instead to make sure that all subfolders are included:

```
addpath(genpath(fullfile(pwd, '..', 'Matlab_Routines')));
```

Data Download & Repository Setup

Before proceeding with the numerical examples, ensure you have obtained the repository code and dataset:

1. **Clone the Repository:** Clone or download the codebase from GitHub:

https:
[//github.com/saddamhijazi/Codes_Manuscript_DataDrivenMOR_for_WavePbs](https://github.com/saddamhijazi/Codes_Manuscript_DataDrivenMOR_for_WavePbs)

2. **Download the Dataset:** Download the pre-computed dataset from Zenodo [?].

3. **Organize Files:** Move the contents of each dataset folder into its corresponding directory within the cloned repository. For example:

`Data_.../NumericalExample4.1/ → Codes_.../NumericalExample4.1/`

Numerical Example 4.1: Damaged Wave Equation

This section outlines the steps required to reproduce the figures for the first numerical example in Section 4.1 (the damaged wave equation). First, navigate to the folder named `NumericalExample4.1`.

Important Note: MATLAB PDE Toolbox Compatibility

Please note that the Full Order Model (FOM) wave equation was solved using the Partial Differential Equation (PDE) Toolbox from **MATLAB R2023b**. Slight numerical discrepancies may occur if executed on different MATLAB releases. If you are using a different MATLAB version, we recommend using the provided pre-computed snapshot files. To do so, ensure `Snapshots.mat` and `U_multi_input.mat` are in your working directory, then set the variable `compute_sol = 0` in the MATLAB code to load pre-computed snapshots directly instead of re-running the PDE solver.

Step 1: Generate POD Basis and Snapshots

To generate Figures 1 and 2, execute the MATLAB script:

```
generatePODBasisMatrix.m
```

This script also generates the snapshot matrices, the POD basis, and scaling factors required by subsequent routines.

Step 2: Continuous Time Operator Inference (OpInf)

Next, compute the Continuous Time OpInf matrices using Python and TensorFlow across reduced dimensions $r \in \{10, 20, 30, \dots, 100\}$. Run the following command in your terminal:

```
python3 -u run_different_opinf_problems.py > runs.log
```

Alternative: To skip the model training step, rename the folder `OpInf_Matrices_precomputed` to `OpInf_Matrices`.

Step 3: Run Model Reduction Solvers

To evaluate POD and Continuous Time OpInf results:

1. Open `different_runs.m` and set the target script:

```
scriptFile = 'POD_OpInf_Results.m';
```

2. Run `different_runs.m` to solve the POD and OpInf systems across all reduced dimensions iteratively.
3. Repeat the execution for Dynamic Mode Decomposition (DMD) by changing the target script to:

```
scriptFile = 'DMD_Results.m';
```

4. Repeat for discrete time OpInf (dOpInf) by setting:

```
scriptFile = 'dOpInf_Results.m';
```

Generated File Structure

After completing the runs, output data will be structured as follows:

1. Method-Specific Results

Files matching the naming pattern below are created for reduced dimensions $r \in \{10, 20, \dots, 100\}$:

- FOM_vs_PODGROM_Results_{r}.mat
- FOM_vs_OpInf_Results_{r}.mat
- FOM_vs_DMD_Results_{r}.mat
- FOM_vs_dOpInf_Results_{r}.mat

Each file contains the time vector, Full Order Model (FOM) sensor signals, reduced-order approximations at the sensor location, and spatial relative errors over time. Standard and optimal DMD results are merged within the DMD files; similarly, dOpInf variants (with and without re-projection) are merged in the dOpInf files.

2. Global Summary Files

The following aggregate files contain the sensor relative errors evaluated across all values of r :

- PODROM_results_different_r.mat
- DMD_Epsilon_s_vs_r.mat
- DMD_Opt_Epsilon_s_vs_r.mat
- OpInf_results_different_r.mat
- dOpInf_results_different_r.mat
- dOpInfReproj_results_different_r.mat

Note: For dOpInf variants, these aggregate files also include the relative operator errors and condition numbers of the matrices inverted during resolution of the linear systems.

Step 4: Generate Figures 3–7

To generate Figures 4–8, execute the MATLAB script:

```
figures.m
```

The resulting figures will open sequentially as MATLAB figures numbered 1 through 9. These correspond to the subplots in Figures 4–8 in the manuscript, ordered left panel first, then right panel.

Step 5: Generate Time Extrapolation Figures (Figure 8)

Reminder: As mentioned in the earlier compatibility note, different MATLAB releases may produce slight variations when re-running the PDE solver. To ensure exact reproducibility, pre-computed snapshot files are provided for this case as well.

Before running the evaluation, ensure you have downloaded the following folders into your working environment:

- **T120:** Contains the full snapshot matrices required for the extrapolation scenario.
- **T40:** Contains the snapshot data used to solve the discrete time OpInf (dOpInf) problem.

The folder **Extrapolation_Case** includes the pre-computed snapshot file **Snapshots.mat**, which stores the full snapshots alongside the mass and stiffness matrices for this specific setup. If you are using a MATLAB version other than R2023b, ensure `compute_sol = 0` in `main.m` to load **Snapshots.mat** directly and also download the folders **T40** and **T120** and place them in the directory **Extrapolation_Case**.

To generate the final figures for the time extrapolation case, navigate to the **Extrapolation_Case** directory and execute:

```
main.m
```

The resulting plots correspond to the subplots in Figure 9 of the manuscript.

Numerical Example 4.2: Damaged Fiber Metal Laminate

This section outlines the steps required to reproduce the figures for the second numerical example in Section 4.2 (the damaged fiber metal laminate). First, navigate to the folder named **NumericalExample4.2**.

Step 1: Generate Singular Value Decay, Cumulative Sum, and DMD Results

To generate Figure 9 (right panel) and Figures 10, 11, and 12, execute the MATLAB script:

```
figures_part1.m
```

Step 2: Generate Operator Inference Results (Figures 14 and 15)

To generate the left and right panels of Figure 13, modify the target reduced dimension r in `main.m`:

1. Open `main.m` and locate the line defining the reduced dimension:

```
r = ...
```

2. Set $r = 50$ and run `main.m` to generate the left panel of Figure 14.

3. Next, set $r = 80$ and run `main.m` again. The first output plot corresponds to the right panel of Figure 13. The second output plot corresponds to Figure 14 (right panel). *Note: The corresponding figure for $r = 50$ is not included in the manuscript.*

To generate the aggregated data for Figure 14 (left panel), run the driver script:

```
different_runs.m
```

This script computes the OpInf displacement approximations at the sensor location and saves them to `OpInf_results_different_r.mat`.

Finally, execute the following script to generate Figure 14 (left panel):

```
Ex2_Figure_Sensor_Error_vs_r.m
```

Numerical Example 4.3: Nonlinear Aluminium Model

This section outlines the steps required to reproduce the figures for the third numerical example in Section 4.3 (the nonlinear aluminium model). First, navigate to the folder named `NumericalExample4.3`.

Step 1: Generate Singular Value Decay and Cumulative Sum Figures

To generate Figure 16, execute the MATLAB script:

```
Ex3_SV_decay_and_cumsum_Figures.m
```

Step 2: Generate DMD Figures

To generate Figure 17 and Figure 18 (right panel), execute the MATLAB script:

```
NonlinearModel_mrdmd.m
```

The first output window corresponds to Figure 17, while the second output window corresponds to the right panel of Figure 18.

To generate the left panel of Figure 18, execute:

```
NonlinearModel_optimalDMD.m
```

Finally, to generate Figure 19, execute:

```
NonlinearDamagedModel_mrdmd.m
```

References